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A Scalable Explainable Deep Learning Framework for Predictive Analysis and Interpretability of CO_2 Emissions Patterns

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ABSTRACT

For environmental monitoring and making decisions that are good for the environment, it is important to be able to accurately anticipate CO_2 emissions. This paper presents an interpretable deep learning framework utilizing a BiLSTM network combined with SHAP for CO_2 prediction. Using a sliding window technique, CO_2 time-series data are reformed to show how things change over time. To stop data leaking, chronological data separation and min-max normalization are used. The proposed BiLSTM model achieved the best result. It has an RMSE of 8.58×10^5, an MAE of 1.19×10^5, a R^2 score of 0.99, and a MAPE of 0.0073% on an unseen dataset. Global and local explanations based on SHAP give complementing feature-level insights using a distinct explainability model trained on engineered country, sector and temporal attributes.

1 Introduction

Carbon dioxide (CO_2) emissions are becoming a global concern because they have a big effect on climate change, global warming, and the long-term health of the planet[1]. The ongoing rise in CO_2 levels, mostly fueled by industrial expansion, transportation, energy use, and urbanization, presents significant threats to ecosystems and human health[2]. Governments, environmental groups, and researchers need to have the ability to accurately predict CO_2 emissions in order to make good policies, come up with plans for sustainable growth, and keep an eye on progress toward climate targets like the Paris Agreement. But it's still hard to anticipate CO_2 emissions because the data is very non-linear, dependent on many things, and often shows long-term temporal correlations[3]. Additionally, CO_2 emission patterns change greatly between regions and periods because of differences in economic activities, energy systems, technological progress, and regulatory frameworks[4]. This makes it even harder to make accurate predictions. Changes in the seasons, rapid changes in policy, and unexpected social and economic events can also cause fast changes in emission trends, which makes standard prediction methods less useful[5]. Therefore, resilient and flexible forecasting models are essential to effectively represent intricate temporal dynamics and yield dependable insights for long-term environmental planning and climate mitigation efforts.

Deep learning techniques have proven highly effective in modeling and forecasting intricate time-series data by identifying nonlinear correlations and enduring temporal dependencies[6]. Bidirectional Long Short-Term Memory (BiLSTM) networks have performed exceptionally well among them. BiLSTM models learn from both the past and the future, which lets them find deeper patterns and temporal correlations than regular LSTMs[7]. BiLSTM is a good choice for CO_2 forecasting because it can capture long-term temporal dependencies in sequential emission data. A lot of individuals think of deep learning models as black boxes because it is hard to understand how they make decisions even though they are very effective at predicting things[8]. The lack of interpretability limits their practical use in environmental and policy-related applications, where understanding the reasons behind predictions is as crucial as attaining high accuracy.

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To find a solution around this problem, Explainable Artificial Intelligence (XAI) has become more and more important. XAI approaches try to make AI models easier to understand, more transparent, and more reliable. SHAP (SHapley Additive exPlanations) is one of the most popular XAI approaches because it gives both global interpretability, which shows how important features are to the model as a whole, and local interpretability, which shows how each feature affects individual predictions[9]. Combining SHAP with prediction models to analyze the influence of engineered country, sector and temporal parameters on CO_2 emission behavior

In this study, we present a BiLSTM-based framework for predicting CO_2 emissions, enhanced by SHAP-driven explainability. The model is made to make accurate forecasts while also being clear and easy to understand, which makes it better for use in the actual world for climatic and environmental purposes. BiLSTM and SHAP work together to close the gap between accurate predictions and insights that people can understand. The work helps policymakers and environmental specialists make better decisions by helping them find the main causes of emissions and comprehend how changes in input features will affect future CO_2 levels. The main purpose of this research is to help achieve sustainable development goals by finding smart, understandable, and future-proof ways to anticipate CO_2 levels.

1.1 Motivation

The amount of CO_2 in the air is rising quickly, which is directly causing climate change. To make better decisions about the environment and plan for the future, we need to be able to accurately anticipate future CO_2 levels. Conventional models fail to capture the intricate patterns in emission data, whereas deep learning approaches are characterized by a lack of transparency. So, we need a model that is both correct and easy to understand. This is why BiLSTM is used for prediction and SHAP is used for interpretation. They help experts determine what factors are most important in causing CO_2 emissions.

1.2 Contributions

The key contributions of this work are as follows:

- Developed a deep learning model that uses Bidirectional LSTM to make reliable predictions about future CO_2 emissions trends based on complex time-series data. The proposed BiLSTM architecture provides an effective task of capturing long-term temporal dependencies and non-linear patterns in emission data. This makes it feasible to make more accurate predictions than with traditional machine learning and unidirectional recurrent models.
- SHAP is used to give global and local interpretability through a supplemental explainability pipeline that analyzes engineered country, sector, and temporal aspects associated with CO_2 emission behavior.
- The BiLSTM and SHAP framework together make models more transparent and help people make good decisions about the environment and policy. The approach addresses the problem of deep learning models being black boxes by combining strong predictive performance with explainability. This builds trust and makes it more likely that AI-driven technologies will be used in real-world environmental and policy-making situations.
- The proposed approach shows the most important things that cause CO_2 emissions, which helps us understand them better and come up with more effective ways to reduce them. The framework gives useful information for focused mitigation measures by finding the main elements that cause emissions. These insights help people make smart choices about long-term climate planning and sustainable development.

While BiLSTM and SHAP are established techniques independently used for predictive modeling and explainability, they the contribution of this work is not intended to introduce an entirely new neural architecture from those techniques alone. This paper develops an interpretable forecasting framework that integrates a temporal consistent preprocessing step, a temporal model built on a stack of BiLSTMs, and dual-level SHAP explainability, all within a unified pipeline, thus allowing for reliable prediction of CO_2 levels.

2 Literature Survey

Numerous research studies in the literature have examined the issue of predicting CO_2 emissions through statistical, machine learning, and deep learning methodologies. XAI approaches are being used increasingly in several fields in recent years to make complex prediction systems more transparent, understandable, and reliable.

Meng et al.[10] examined worldwide CO_2 emission forecasts by analyzing the effects of the COVID-19 epidemic through SARIMAX models. There were four models made for distinct stages of the pandemic: before COVID, when COVID started, when COVID was changing, and after COVID. They used Mean Absolute Percentage Error (MAPE) to test the models. The post-COVID model achieved the best performance. Additionally, Ajala et al.[11] used statistical, machine learning, and deep learning models to look at how well daily CO_2 emissions might be predicted in areas with a lot of pollution. Their results

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indicated that machine learning and deep approaches surpassed statistical models, with ensemble-based machine learning techniques attaining similar accuracy to deep learning models at reduced computing expense, rendering them appropriate for short-term CO_2 prediction. In addition, Giannelos et al.[12] studied long-term CO_2 emission forecasting from the building industry through several machine learning methodologies, including Linear Regression, ARIMA, shallow neural networks, and deep neural networks. Their study, conducted in several global areas, illustrated the efficacy of ML-based models for predicting building-related CO_2 emissions till 2050. , Additionally, Kumari et al.[13] examined India's CO_2 emission forecasting utilizing univariate time-series data from 1980 to 2019. They looked at several kinds of statistical, machine learning, and deep learning models, such as ARIMA, SARIMAX, Holt-Winters, Linear Regression, Random Forest, and LSTM. The findings showed that the LSTM model had the least amount of prediction error, which shows that deep learning works well for predicting CO_2 emissions over a lengthy period of time. Additionally, Nguyen et al.[14] did a thorough evaluation of ML and DL methods for predicting CO_2 emissions. They looked at both supervised and unsupervised models and used metrics like R^2, MAE, and RMSE to do so. Their study highlights the robust predictive power of deep learning, artificial neural networks (ANN), and support vector machine (SVM) models, while also emphasizing the efficacy of hybrid and data-driven methodologies for precise carbon emission forecasting. In addition, Li et al.[15] investigated near-real-time daily CO_2 emission forecasting in China utilizing univariate time-series data, contrasting statistical models (GM(1,1), ARIMA, and SARIMAX) with machine learning models (ANN, RF, and LSTM). The findings indicated that machine learning models consistently surpassed statistical methods, with the LSTM model demonstrating predictive performance in RMSE, MAE, MAPE, and R^2. Additionally, Stefano et al.[16] proposed a detailed approach for estimating urban CO_2 emissions that makes use of open-access socio-economic and city service data. ML models, especially XGBoost and Graph Convolutional Networks (GCN), were quite good at making predictions (MAPE = 8%). The study utilized Explainable AI to pinpoint significant urban elements affecting emissions, illustrating the efficacy of data-driven and interpretable models for sustainable urban planning and policy development. In addition, Tiwari et al.[17] put forward a DL-based framework that is integrated with XAI. For transparency and trust, XAI techniques were used with SHAP. These approaches gave both global and local interpretations through summary and waterfall plots. Similarly, Volkan et al.[18] estimated hourly CO_2 emissions from maritime transportation by employing Random Forest and Extreme Gradient Boosting models, which were trained on operational and structural ship data. Both models were very accurate ($R^2 = 0.92$), although XGBoost had a lower RMSE and MAE than Random Forest. SHAP-based explainability showed that gross tonnage, speed, and length were the most important parameters. A model with fewer features did just as well, showing that explainable ML models are useful for following rules and managing maritime emissions in a way that is good for the environment. Additionally, Tiwari et al.[19] proposed an explainable BiLSTM framework that works with SHAP. This shows how XAI provides global and local explanations for deep learning-based time-series forecasting, which is very useful for predicting CO_2 emissions in a way that is intuitive and understandable. Additionally, Dwivedi et al.[20] emphasized that XAI increases the clarity and understandability of machine learning models, which builds user trust and makes it easier to apply in important areas. Their study categorized current XAI methodologies and frameworks to facilitate the choice of suitable explainability strategies at various phases of the machine learning pipeline. In addition, Tervo et al.[21] developed a model for how to use waste heat from data centers in a district heating network in Finland. They showed that using waste heat can lower CO_2 emissions and fuel costs in a cost-effective way while also helping the local economy and following the law. Additionally, Liu et al.[22] developed a machine learning-based QSPR model that can be understood to predict how well CO_2 dissolves in aqueous amine solutions. SHAP analysis showed how important molecular descriptors, pressure, temperature, and solution concentration are, making it easier to model CO_2 capture in a clear way.

Although previous research has yielded encouraging outcomes in CO_2 emission forecasting, they predominantly emphasize accuracy and often lack interpretability. The application of XAI techniques in deep learning–based CO_2 forecasting is still limited. This drives the proposed BiLSTM–SHAP paradigm, which seeks to deliver greater accuracy in prediction and enhance the transparency as well as trust, which helps in decision-making.

3 Proposed Scheme

The proposed work introduces an interpretable deep learning framework for predicting CO_2 emissions via a BiLSTM network. Figure 2 shows the overall pipeline for the proposed work. The architecture of the proposed framework consists of several sequential stages, each responsible for a specific task in the CO_2 prediction and explanation process. The methodology consists of two complementary parts: (1) a univariate BiLSTM forecasting pipeline that leverages historical CO_2 sequences and (2) an additional explainability pipeline that leverages engineered country, sector, and temporal features. Every part of the architecture is made up of numerous levels, and each layer has different tasks to do.

3.1 BiLSTM-Based CO_2 Prediction Module

Figure 1 shows the detailed design of the proposed stacked BiLSTM-based CO_2 prediction module. The model uses a sliding window technique, where a sequence of the last 10 consecutive CO_2 observations $(x_{t-9}, x_{t-8}, \ldots, x_t)$, each with a size of 10×1,

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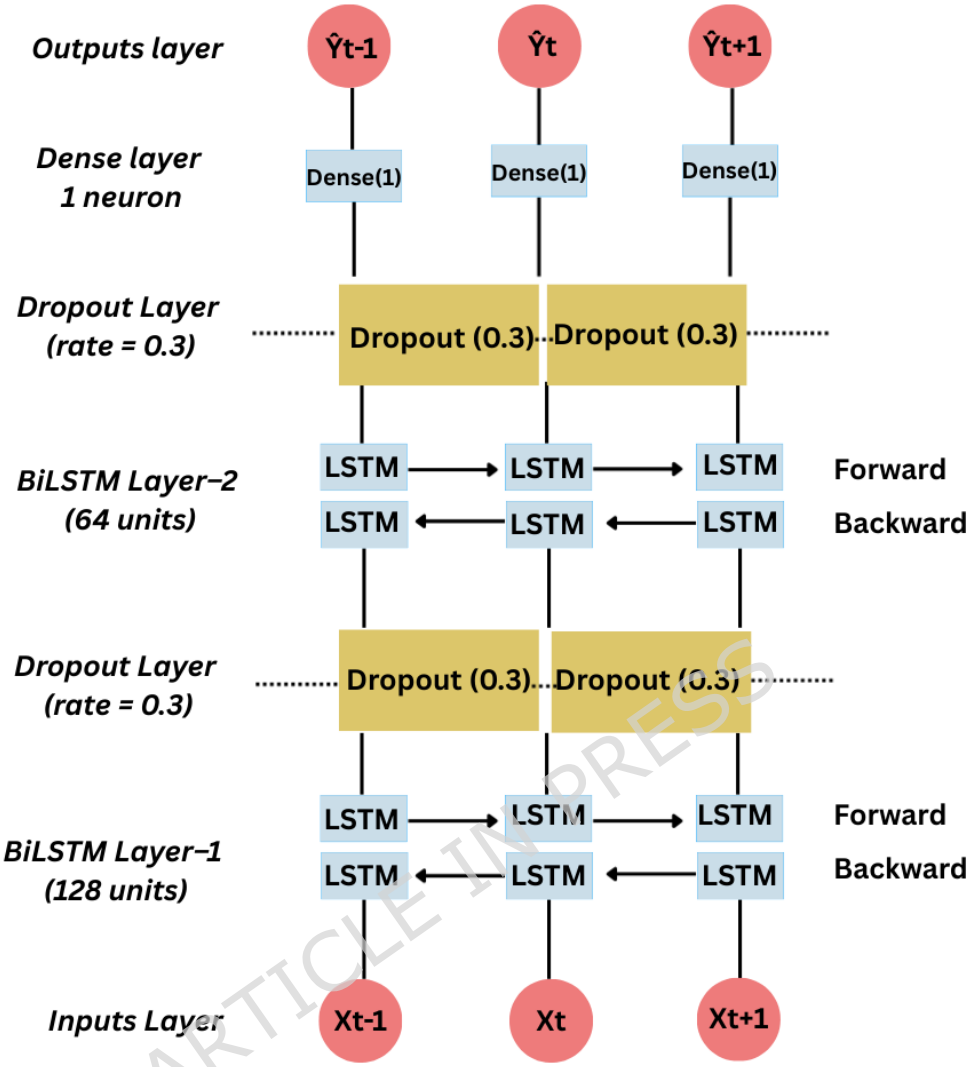


Figure 1. Stacked BiLSTM architecture for CO_2 prediction

is utilized as the input feature vector. A sliding window length of 10 observations was used to provide enough historical information to learn the temporal dependencies in the CO_2 emission sequence while keeping the input dimensionality within reasonable bounds. Very small window sizes may miss short- and medium-term temporal patterns, while overly large windows might contribute redundant information and increase computational complexity and potentially impact model generalization. For the proposed BiLSTM forecasting framework, a window size of 10 was adopted as a practical trade-off between the benefits of the magnitude of the temporal information retained and the computing efficiency. The first BiLSTM layer, which has 128 hidden units, analyzes the input sequence in both forward and backward directions. This allows it to capture past and future contextual dependencies within the input window. To avoid overfitting, a dropout layer with a rate of 0.3 is added after this layer. This randomly deactivates the neurons during training. After that, the extracted features are transferred to a second stacked BiLSTM layer with 64 hidden units. This layer models higher-level sequential patterns to make the learned temporal representations even better. After the second BiLSTM layer, another dropout layer with a rate of 0.3 is added to make things even more regular. The final output of the stacked BiLSTM layers goes into a fully connected dense layer with one neuron. This neuron makes the predicted CO_2 concentration for the next time step $\hat{y}_{t+1}$. This many-to-one predictive approach allows the model to predict future CO_2 values based on historical temporal patterns. The proposed forecasting architecture is implemented as a univariate forecasting framework, where only the historical CO_2 data are fed into the BiLSTM model. The proposed architecture of the BiLSTM model was designed according to the common configurations used across deep

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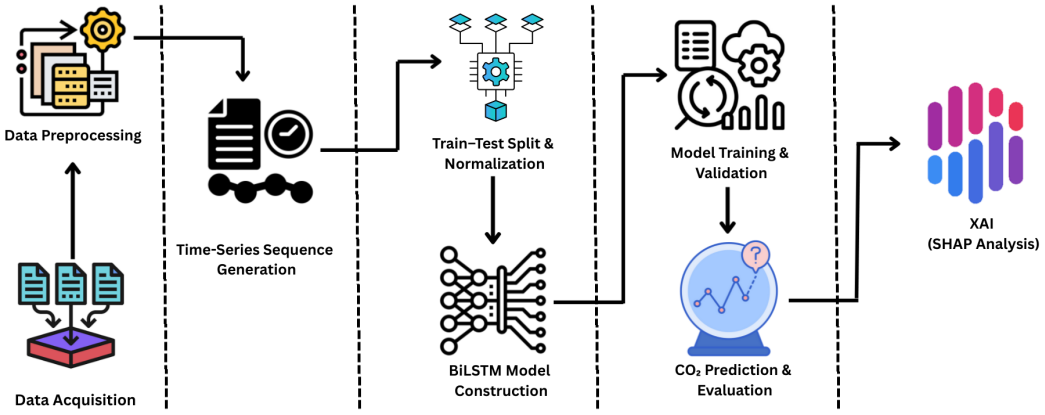


Figure 2. Proposed BiLSTM-XAI-based CO_2 prediction framework

learning-based time series forecasting studies and through empirical experimentation during the model development process. The first layer of the proposed BiLSTM model consists of 128 hidden units to model higher-order temporal dependencies contained within the input sequence. The second layer consists of 64 hidden units to further process previously learned temporal representations while reducing model complexity. A 0.3 dropout rate was employed after each of the two BiLSTM layers to reduce overfitting and provide improved generalization capabilities. During the model development there were numerous preliminary configurations that were evaluated; the selected architecture showed the most stable convergence, least amount of validation loss and the best predictive performance relative to other configurations. Therefore, this architecture will be used as the basis for all subsequent experiments.

3.2 Justification for Using BiLSTM

Bidirectional Long Short-Term Memory (BiLSTM) is employed as the core prediction model due to its ability to learn temporal dependencies in time-series data. CO_2 data exhibits long-term trends, seasonal variations, and short-term fluctuations caused by industrial activity, transportation, and environmental factors. Traditional machine learning models and unidirectional LSTM networks process data only in the forward direction, which may limit their ability to fully capture contextual temporal information. BiLSTM addresses this limitation by processing the input sequence in both forward and backward directions, enabling the model to learn information from past as well as future time steps. This dual learning mechanism allows BiLSTM to capture hidden temporal patterns more effectively than standard LSTM. The key reasons for selecting BiLSTM in this work are: To enhance temporal feature learning, BiLSTM is used; it captures both historical and future dependencies with CO_2 time-series data. To improve accuracy prediction Bidirectional processing improves forecasting performance for nonlinear and nonstationary data. To handling robust long-term dependencies the LSTM memory cell structure mitigates the vanishing gradient problem. It is Suitable for environmental data CO_2 concentration changes are influenced by cumulative and sequential factors. The BiLSTM architecture was not the only predictive model investigated in this research study. Other predictive methodologies like classical ML models, Support Vector Regression (SVR) and Gradient Boosting were also evaluated to use as a baseline for comparative assessment. The reason why BiLSTM became the main predictive method is due to its ability to model temporal dependencies and nonlinear relationships for capturing temporal dependencies (bidirectionally) and for capturing the nonlinearities in the CO_2 emissions time series dataset. The comparative results provided in this evaluation support this decision. SHAP was also included as an explainability mechanism in this framework but is not uniquely considered to be a prediction mechanism. Even though SHAP can be used across many ML and DL models, it was important that a form of SHAP be utilized in this study since it can help produce both global and local interpretations for BiLSTM predictions. As a result, the contribution of this work will provide improved transparency, interpretability, and trustworthiness; the prediction will be possible via the combination of accurate CO_2 forecasting via the use of BiLSTM and explainability via SHAP.

3.3 CO_2 Prediction Framework

The algorithm 1 shows how the proposed BiLSTM-based CO_2 prediction model works, including SHAP for global and local explainability. Table 1 presents the key notations used in Algorithm 1 for reference. The first step is to load the CO_2 time-series dataset. Based on this, the target variable Y (CO_2 concentration) and the features of the input X are separated (Algorithm 1: lines 1–2). The CO_2 time-series data is restructured into a supervised learning problem by employing a sliding window of 10 preceding observations to forecast the subsequent emission value (see Eq. 1); Algorithm 1: line 3). The window length was

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set to 10 observations to ensure that there was enough historical information available for modeling temporal dependencies, without unnecessarily increasing sequence length and model complexity. To maintain temporal dependencies and prevent data leaking, the dataset is split into two parts: 80% for training and 20% for testing, represented as $X_{\text{train}}, X_{\text{test}}, Y_{\text{train}}, Y_{\text{test}}$ Algorithm 1: line 4. Then, Min-Max normalization is applied to ensure numerical stability, where scaling parameters are only learned from the training data (see Eq. 2); (Algorithm 1: lines 5–6). This ensures numerical stability and uniform scaling during model training and inference. Subsequently, the BiLSTM model M is constructed with a sequential architecture (Algorithm 1: line 7). The proposed model consists of two stacked BiLSTM layers, one with 128 hidden units and the other with 64 (Algorithm 1: lines 8 and 10). Eq. 3 mathematically defines the bidirectional hidden state representation of the BiLSTM. After each layer, there is dropout regularization with a rate of 0.3 to stop overfitting (Algorithm 1: lines 9 and 11). Finally, there is a fully connected dense layer for regression output as shown in Eq. 4 (Algorithm 1: line 12) for generating the continuous CO_2 prediction output. The Adam optimizer trains the network, and the mean squared error is the loss function (Algorithm 1: line 13). The network is trained for a maximum of 60 epochs with a batch size of 16 while early stopping based on validation loss is used to make sure that the network converges steadily (Algorithm 1: line 14). After completion of training, the trained model is employed to predict CO_2 values on the test dataset (Algorithm 1: line 15). The anticipated and actual values are subsequently inverse-transformed to their original scale for significant evaluation (Algorithm 1: line 16). We use RMSE, MAE, the coefficient of determination (R^2), and the mean absolute percentage error (MAPE) as defined to measure how well a model works on unseen test data (Algorithm 1: line 17). To ensure proper time series forecasting and eliminate the risk of data leakage from the dataset, all data is split chronologically and without shuffling into training and testing (and validation) datasets. All preprocessing steps (including Min-Max normalization) are fitted to the training data only and then applied to the test data. Future value predictions use the sliding-window process, where only historical observations can be used

to forecast future values, ensuring that the time series forecasting setup is fully causal and consistent with real-world time series forecasting situations. To enhance interpretability and provide feature-level explanations, SHAP analysis is performed using a distinct baseline machine learning model trained on synthetic country, sector and temporal variables. The proposed BiLSTM model is used for temporal CO_2 predictions while SHAP explanations provide complementary dataset-level insights on feature contributions and emission behavior.

For the explainability analysis, engineered features are created from country, sector and temporal information and used to train a different baseline machine learning model (Algorithm 1: lines 18-19). Prior to model training, the categorical variables country and sector are converted into integer-encoded numerical values. The features generated are country_enc and sector_enc. The encoded characteristics and temporal aspects are employed just in the baseline explainability model, and are not given as an input to the proposed BiLSTM forecasting architecture. This baseline model is then used to initialize a SHAP explainer to compute feature attributions (Algorithm 1: lines 20-21). SHAP summary visualizations are used to achieve global interpretability, examining the overall feature relevance and contribution patterns for CO_2 emissions (Algorithm 1: line 22). Local interpretability is accomplished by selecting representative samples and generating force plots and waterfall plots for visualization of individual feature contributions toward prediction behavior (Algorithm 1: lines 23–25). This complementary explainability analysis gives further insights into emission-related feature interactions while maintaining the proposed BiLSTM architecture as the main forecasting model.

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$$X_t = \{y_{t-T+1}, y_{t-T+2}, \ldots, y_t\}, \quad Y_t = y_{t+1} \tag{1}$$

where X_t denotes the sliding-window input sequence of length T, Y_t denotes the next CO_2 value to be predicted and y_t denotes the CO_2 observation at time step t. In this work, $T = 10$.

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \tag{2}$$

where x is the original feature value, x' denotes the normalized value, $x_{\min}$ and $x_{\max}$ represents the minimum and maximum values computed from the training data.

$$h_t = \left[\overrightarrow{h_t} \parallel \overleftarrow{h_t}\right] \tag{3}$$

where $\overrightarrow{h_t}$ and $\overleftarrow{h_t}$ denote the forward and backward hidden states of the BiLSTM at time step t, respectively, and $\parallel$ represents the concatenation operation.

$$\hat{y} = Wh + b \tag{4}$$

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where $\hat{y}$ is the predicted CO_2 value, W denotes the weight matrix, h represents the final BiLSTM hidden representation, and b is the bias term.

$$f(x) = \phi_0 + \sum_{i=1}^{M} \phi_i \tag{5}$$

where $f(x)$ denotes the model output, ϕ_0 represents the base value, ϕ_i indicates the SHAP contribution of the i-th feature, and M is the total number of features.

Algorithm 1 effectively combines accurate deep learning-driven time-series prediction with strong explainability, facilitating both superior CO_2 prediction and comprehensible decision-making. To enhance the clarity, transparency, and dependability of the proposed BiLSTM-based CO_2 prediction model, Explainable Artificial Intelligence (XAI) is integrated via SHAP. In the baseline explainability model, we employ SHAP to measure the contribution of engineered country, sector, and temporal factors to the CO_2 prediction behavior. SHAP summary graphs show the global importance of features, indicating both the magnitude and direction of their influence on CO_2 emissions. This is how global model behavior is analyzed. To explore the significance of features on a global scale, SHAP summary plots are used to demonstrate the size and the direction of the influence of features on the CO_2 predictions.

SHAP force plots and waterfall plots make local interpretability possible by breaking down individual predictions into positive and negative feature contributions based on the model's base value. Force plots highlight the directional impact of each feature, whereas waterfall plots offer a cumulative and incremental representation of how various feature impacts aggregate to yield the final prediction. Ordering local explanations by output magnitude allows for a clear observation of the change in feature influence across different emission levels. The SHAP output values quantitatively show how designed country, sector and temporal variables affect prediction behavior, hence increasing transparency and interpretability.

Notation	Description
D	Input dataset
X	Sliding-window sequences of historical CO_2 observations
Y	Target variable (CO_2 concentration)
$X_{\text{train}}, X_{\text{test}}$	Training and testing feature sets
$Y_{\text{train}}, Y_{\text{test}}$	Training and testing target values
$F_{\text{engineered}}$	Engineered features (country, sector, temporal attributes)
M	Stacked BiLSTM model
BiLSTM	Bidirectional Long Short-Term Memory network
H	Training history
Y_{pred}	Predicted CO_2 values (normalized)
$Y_{\text{test_inv}}, Y_{\text{pred_inv}}$	Actual and predicted CO_2 values (original scale)
DL	Deep Learning
DNN	Deep Neural Network
XAI	Explainable Artificial Intelligence
SHAP	SHapley Additive exPlanations
BaselineModel	Classical ML model used for SHAP explainability
E_{global}	SHAP explainer (global, for BaselineModel)
S_{global}	SHAP values (global)
x_{sample}	Selected sample for local explanation
S_{local}	SHAP values (local)

Table 1. Key Notations

4 Performance evaluation

This section presents a performance evaluation of the proposed BiLSTM-based CO_2 prediction model utilizing SHAP for XAI-based interpretability.

4.1 Environment setup for experiment

The computational experiments of this research have been done on a computer device with a 2.40 GHz Intel(R) Core(TM) i7-6300U CPU, 16GB of RAM, along with 512GB of SSD. The device also has an NVIDIA GeForce 940MX GPU, working on a 64-bit architecture. The whole experimental code has been performed using the libraries, including TensorFlow and Keras, simplifying the creation and training of the proposed model.

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Algorithm 1 CO_2 Prediction Using Stacked BiLSTM with SHAP Explainability

$D \leftarrow$ Load Dataset
$Y \leftarrow D[\text{CO2}]$
$[X, Y] \leftarrow \text{CreateTimeSeriesSequences}(Y, \text{time_step} = 10)$
$[X_{\text{train}}, X_{\text{test}}, Y_{\text{train}}, Y_{\text{test}}] \leftarrow \text{SplitTrainTest}(X, Y, 0.8)$
Normalize training and testing data
Build stacked BiLSTM model M
Add BiLSTM(128)
Add Dropout(0.3)
Add BiLSTM(64)
Add Dropout(0.3)
Add Dense(1)
Compile model using Adam optimizer and MSE loss
Train model M using EarlyStopping
$Y_{\text{pred}} \leftarrow M.predict(X_{\text{test}})$
Inverse transform predictions
Compute RMSE, MAE, R^2, and MAPE
Explainability Analysis
Construct engineered features $F_{\text{engineered}}$ (country, sector, temporal attributes)
Train BaselineModel using $F_{\text{engineered}}$
Initialize SHAP explainer:
$E_{\text{global}} \leftarrow \text{SHAP.Explainer}(\text{BaselineModel})$
Compute SHAP values:
$S_{\text{global}} \leftarrow E_{\text{global}}(X_{\text{test}})$
Generate summary plot for global explanations
Select representative sample x_{sample}
Generate force plot for local explanations
Generate waterfall plot for local explanations

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4.2 Dataset Description

The dataset utilized in this work consists of extensive CO_2 emission records gathered from various nations, sectors, and time periods, rendering it appropriate for both temporal modeling and cross-sectional analysis and it is taken from Kaggle[23]. There are 135,408 observations in the dataset, and each one has five attributes. Three of these are categorical features: country, sector, and date. The other two are numerical features: CO_2 emission value and timestamp. The country attribute shows different parts of the world, while the sector attribute shows where emissions come from, like electricity generation and other associated activities. The date column keeps track of emissions data on a daily basis and comes in many date formats, which are standardized during preprocessing. The timestamp feature shows Unix time and keeps the order of observations in time, which makes it possible to model things chronologically. The target variable, value, shows the size of the measured CO_2 emissions and varies greatly in number, which reflects how emissions change in different areas and industries. The dataset is complete because there are no missing values. The temporal continuity and worldwide coverage of the dataset, as well as the large scale temporal observations make it suited for deep learning based CO_2 time series forecasting and further explainability analysis. The categorical variables, such as country and sector, are converted into numerical formats for the explainability analysis pipeline. Specifically, categorical variables are transformed into integer representations to create the country_enc and sector_enc features. The encoded variables are not employed as direct inputs in the proposed BiLSTM forecasting architecture; they are just utilized within the supplementary baseline explainability model.

4.3 Related Studies on CO_2 Prediction

Results of the proposed work approach were compared to those from other researchers in terms of RMSE, MAE, R^2 score, and MAPE. Table 2 shows the comparison of the proposed scheme to other schemes.

1. Meng et al.[10] utilized SARIMAX-based time-series models to predict global CO_2 emissions, considering the effects of COVID-19, and attained a low MAPE of 0.09 in post-COVID scenarios.

2. Ajala et al.[11] showed that ensemble-based machine learning models surpass statistical methods in daily CO_2 prediction, with high prediction accuracy (R^2up to 0.93), while incurring lower computing costs than deep learning, thereby rendering them appropriate for short-term emission forecasting.

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3. Giannelos et al.[12] showed that multivariate ML and DNN are better than ARIMA for predicting long-term CO_2 emissions from buildings. This shows how important nonlinear models are for studies on decarbonization in specific areas.

4. Kumari et al.[13] demonstrated that deep learning-based LSTM surpasses statistical and machine learning models in predicting India's CO_2 emissions, attaining minimal error rates (MAPE = 3.10%, RMSE = 60.63) and exhibiting robust proficiency in long-term univariate time-series prediction.

5. Li et al.[15] demonstrated that machine learning models surpass statistical methods for near-real-time daily CO_2 emission forecasting in China, with LSTM achieving the lowest reported prediction error ($R^2 = 0.98$, RMSE = 0.0187), highlighting its efficacy in capturing intricate short-term emission dynamics.

All the studies outlined in Table 2 used different types of data, geographical scales, time scales, feature sets and methods of evaluation when testing their prediction models; therefore, while the performance metrics presented are for reference only and shouldn't be interpreted as concrete quantitative comparisons to the results of the proposed model, it would be necessary to replicate the performances of all of the baseline methods on the same data set under the same experimental conditions to statistically compare them; therefore, the discussion focuses on qualitative differences between the effectiveness of various prediction approaches, instead of saying claiming superiority of one approach over another just because of how many metrics were reported.

4.4 Prediction Results

The performance of the proposed BiLSTM model is compared with prior CO_2 prediction studies in Table 2. The comparison is meant to provide qualitative context and not a directly quantitative benchmark, as the research uses various datasets, geographic scope, temporal resolution and evaluation settings. Due to the datasets, spatial dimensions, and temporal resolutions all being different, the results of earlier studies are shown for contextual comparison instead of direct quantitative benchmarking. In the reported study, the SARIMAX-based approach produced a MAPE of 0.09 but other evaluation metrics such as RMSE and MAE were not included. General machine learning and deep learning models used on a daily and regional basis have a R^2 score of 0.932, which means they are somewhat reliable. Sector-specific deep neural network (DNN) techniques show good long-term prediction power, with a MAPE of 1.65. LSTM-based models applied on national and regional datasets have demonstrated robust predictive efficacy in their particular experimental settings. For example, an LSTM model trained on Indian annual CO_2 emission data gets a R^2 score of 0.99 with an MAE of 45.524 and a MAPE of 3.10. Another LSTM model used on daily CO_2 emissions in China gets a R^2 score of 0.98 and a MAPE of 14.82, emphasizing the suitability of recurrent architectures for temporal modeling challenges. Also, Support Vector Regression (SVR) and Gradient Boosting, two classical machine learning baselines, are evaluated on a global scale annually. The SVR model exhibited a root mean squared error (RMSE) of 1.26×10^6, a mean absolute error (MAE) of 8.57×10^5, and a R^2 score of 0.99 in its evaluation context. The Gradient Boosting model exhibited an RMSE of 8.39×10^5, an MAE of 1.12×10^5, and a mean absolute percentage error (MAPE) of 0.0069. These results are provided just for qualitative reference due to variations in datasets, time scales, and experimental conditions across the study. Gradient Boosting also demonstrated promising predictive performance in its experimental environment. It has an RMSE of 8.39×10^5, an MAE of 1.12×10^5, and a low MAPE of 0.0069, which shows that it can make good predictions. The proposed BiLSTM model demonstrates favorable predictive performance in predicting CO_2 emissions on a worldwide annual basis, with a high R^2 score of 0.99 and a low MAPE of 0.0073%. This means that the predicted and actual emissions are very similar. The results demonstrate that the proposed BiLSTM model efficiently captures temporal dependencies in CO_2 emission data.

The large value of both RMSE and MAE in absolute terms is driven mostly by the size of CO_2 emissions including emissions of approximately 10^9. Thus, relative metrics provide a more meaningful understanding of predictive performance. The proposed BiLSTM model achieves a very low MAPE of 0.0073%, which means that it predicts with very little error relative to the target values. Moreover, the predicted performance was significantly improved since the proposed model reduced RMSE by 31.9% and MAE by 86.1% compared with the SVR baseline.

Figure 3 shows how the training and validation loss of the proposed BiLSTM model changed throughout 60 training epochs. The training loss drops quickly from about 1.2×10^{-3} in the first epoch to roughly 5.5×10^{-4} in the second epoch. This shows that the model is learning quickly initially. Between epochs 3 and 20, the training loss continues to decrease at a moderate rate, indicating progressive refinement of the learned temporal features and stable gradient updates. After that, the loss slowly comes together and stabilizes around 4.2×10^{-4} toward the end of the epochs, showing that the network parameters are being steadily optimized. After this phase, the loss continues to decrease and converges to approximately 3.7×10^{-4}, by the end of training, reflecting a steady and consistent optimization process. The validation loss stays low, dropping below 1.0×10^{-4}. The model is not overfitting because the difference between the training and validation losses is not too big. The smooth and stable convergence of both training and validation loss curves indicates that the proposed BiLSTM model exhibits stable convergence behavior and strong generalization capabilities. Overall, the results demonstrate that the BiLSTM architecture and training

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Figure 3. Training and validation loss convergence of the proposed CO_2 prediction model based on BiLSTM

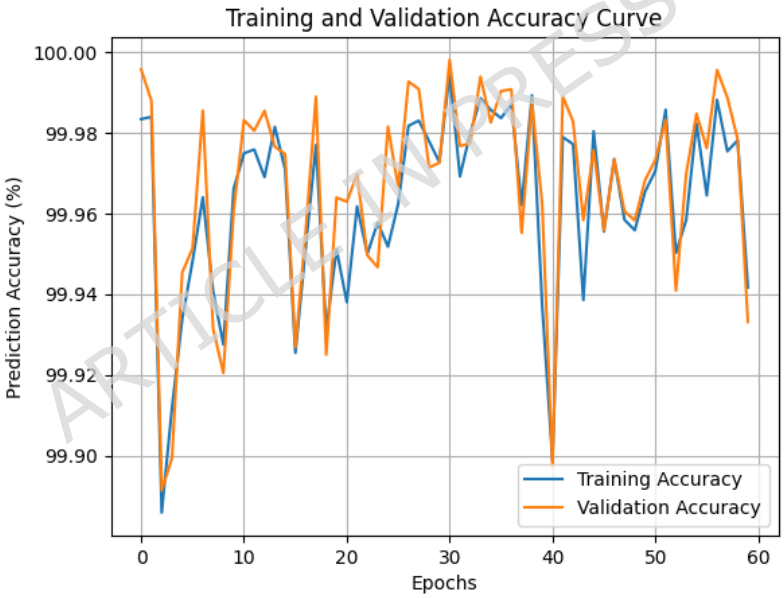


Figure 4. Training and validation accuracy of the proposed BiLSTM model throughout 60 epochs

method work well for predicting CO_2 emissions. Figure 4 indicates that the proposed stacked BiLSTM model consistently has strong predictive performance throughout the entire training process, as demonstrated by the training and

validation accuracy curves. For all 60 epochs, the training and validation accuracy stays between 99.88% and 99.99%, which shows that the learning behavior is consistent and reliable. After an initial learning phase, which continues for the first 5 to 10 epochs, the model rapidly converges, and the accuracy values stay over 99.95%. The highest accuracy seen is close to 99.99%, which shows that the model is quite good at capturing the complex temporal connections that are inherent in CO_2 time-series data. The difference between training and validation accuracy is still very small, consistently less than 0.05%. This shows that the model can generalize well and that overfitting is being controlled well. There are small changes of about ±0.03%–0.07% across epochs. These are normal because of batch-wise optimization in recurrent neural networks and don't mean that the system is unstable. The close alignment of the training and validation curves shows that the dropout regularization (rate = 0.3) performed effectively and shows that the proposed BiLSTM model for CO_2 prediction is strong and

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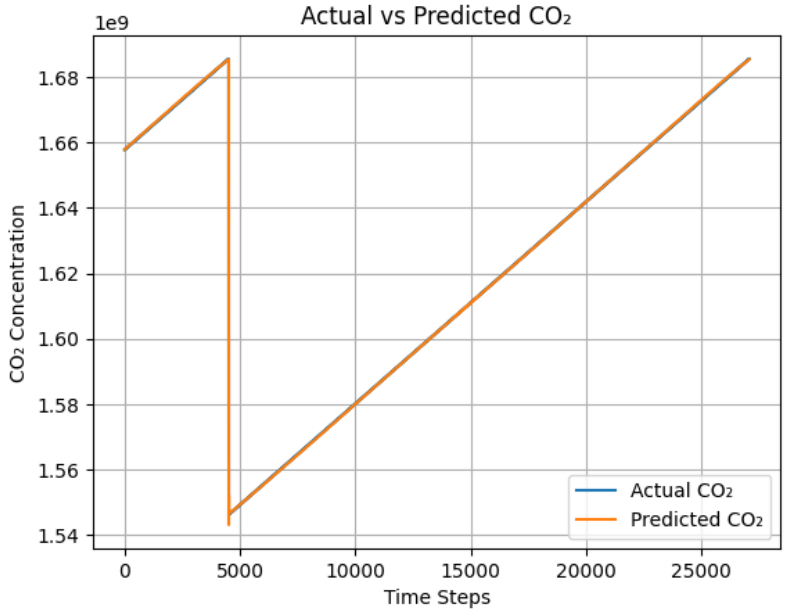


Figure 5. A test set comparison of CO_2 emissions and those predicted by BiLSTM

reliable. Figure 5 shows how actual and predicted CO_2 emissions compare on the unseen dataset. The predicted CO_2 trajectory closely aligns with the actual values during the whole testing period, illustrating the consistent predictive performance of the proposed BiLSTM model. The CO_2 concentration values are between 1.54×10^9 and 1.69×10^9, and the predicted curve is almost exactly the same as the actual curve, which means there is very little difference. The model effectively captures the long-term upward trend and sudden changes in emission levels over time, demonstrating its capacity to learn intricate temporal correlations inherent in CO_2 time-series data. The high coefficient of determination ($R^2$0.99) and low relative error (MAPE is 0.0073%) show that the proposed prediction framework is strong and accurate.

Model Used	**Prediction Scale**	**RMSE**	**MAE**	**R^2 Score**	**MAPE (%)**
SARIMAX[10]	Global / Annual	–	–	–	0.09
ML & DL Model[11]	Regional / Daily	0.247	–	0.93	–
DNN[12]	Building Sector / Long-term	–	–	–	1.65
LSTM[13]	India / Annual	60.635	45.524	0.99	3.10
LSTM[15]	China / Daily	0.0187	0.0140	0.98	14.82
SVR	Global / Annual	1.26×10^6	8.57×10^5	0.99	0.05
Gradient Boosting	Global / Annual	8.39×10^5	1.12×10^5	0.99	0.0069
BiLSTM (Proposed)	**Global / Annual**	8.58×10^5	1.19×10^5	**0.99**	**0.0073**

Table 2. Summary of Reported Results in Previous CO_2 Prediction Studies

4.5 Explainability Results

SHAP-based XAI approaches were employed to conduct complementary interpretability analysis of CO_2 emission behavior, utilizing a distinct baseline machine learning model trained on engineered variables related to nation, sector, and time. The proposed BiLSTM model does temporal forecasting with univariate CO_2 sequences, whereas SHAP analysis offers further feature-level insights into emission patterns. These methods offer local interpretability, elucidating the contribution of features to specific predictions, and global interpretability, which determines the overall significance of features and patterns over the whole dataset.

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Figure 6. Local SHAP explanation of an individual CO_2 emission prediction

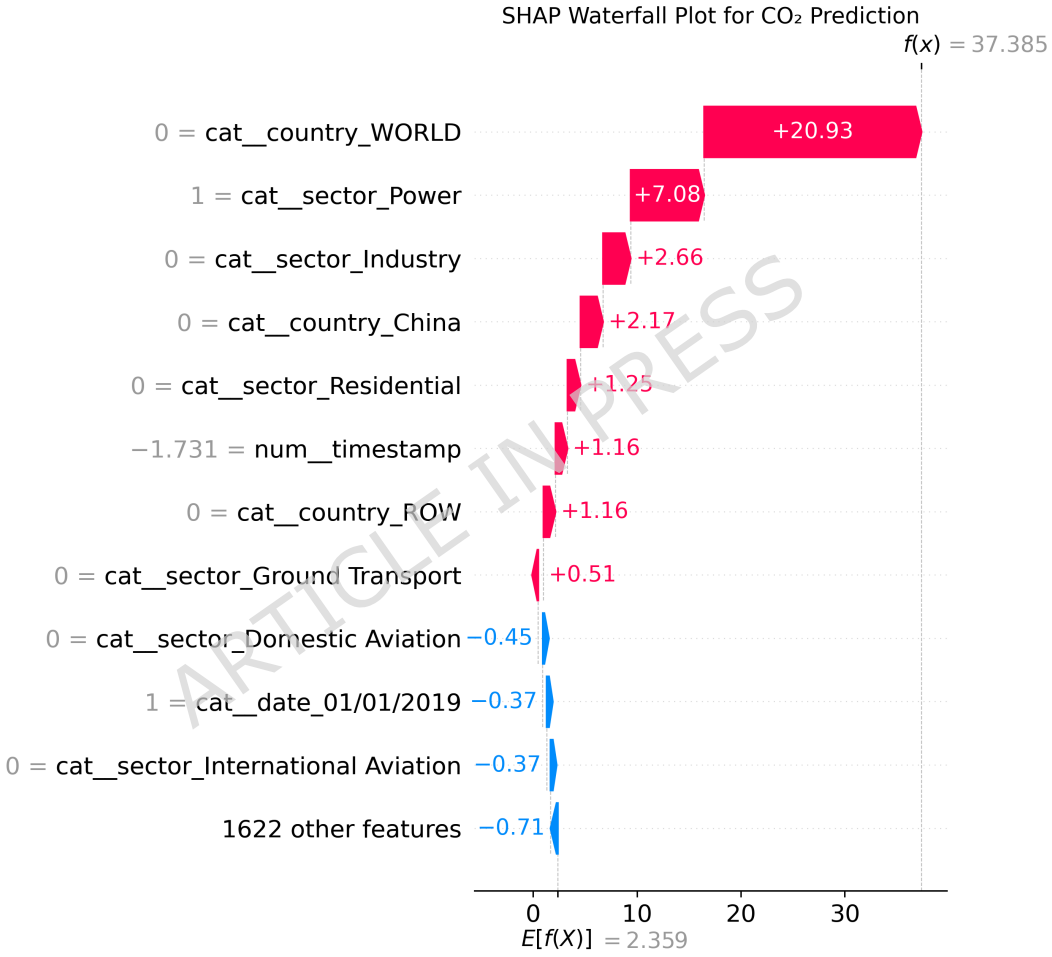


Figure 7. A SHAP waterfall plot demonstrating the contributions of features to one CO_2 prediction. Values before feature names are feature values of the selected sample and bars are their SHAP contributions.

4.5.1 Local Explanations

Figure 6 illustrates a local SHAP force plot explaining an individual CO_2 emission prediction with the baseline explainability model for feature attribution analysis. The figure shows how different input features change the model output from the global baseline value to the final prediction. The baseline value is the average output of the model over the training data. The final prediction, f(x)=2.08, is the sum of both positive and negative feature contributions. Features in red (like year = 2022 and sector_enc = 4.0) make the prediction less accurate by pushing the output toward lower CO_2 values. Features in blue (like country_enc = 2.0 and month = 5.0) make the prediction more accurate by raising the predicted emission level. The magnitude

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and direction of each contribution show how important the feature is for this specific instance. The local explanation shows how the engineered country, sector, and temporal features contribute to the individual prediction behaviors in the explainability model. Figure 7 shows the SHAP waterfall plot that explains one prediction for CO_2 emissions. On the waterfall plot, the values before each feature name are the actual feature values of the sample that is being explained. The number 1 indicates that the relevant category is present (active) in the sample, whereas the value 0 indicates that the corresponding category is absent (inactive). These values give some context information about the observation, whereas the corresponding SHAP values measure the contribution of each feature to increase or decrease the final CO_2 prediction. The baseline number E[f(X)]=2.359 is the average output of the model, and the final prediction, f(x)=37.385, shows that this is a high-emission case. The global country aggregation (+20.93) is the biggest positive factor, followed by the power (+7.08) and industrial (+2.66) sectors. This shows how important energy-intensive industries are. Temporal factors make predictions even higher, showing that emissions are growing with time. On the other hand, aviation-related sectors and some date aspects have a negative effect, which lowers the expected value marginally. The plot shows that the baseline explainability model utilizes country-, sector-, and temporal features to provide interpretable insights into CO_2 emission behavior and feature contributions.

4.5.2 Global Explanations

Figure 8 presents a comprehensive SHAP summary that demonstrates the impact of engineered country, sector, and temporal variables on prediction behavior within the baseline explainability model. The country encoding is the most important feature because it has the biggest range of SHAP values, from about -8 to $+26$. Higher encoded values make the model output much higher. The sector encoding comes in second, with SHAP values ranging from about -8 to $+15$. This shows that emissions vary a lot between different economic sectors. Temporal factors have a smaller effect overall. The month and timestamp have a modest effect, with SHAP magnitudes mostly within ± 3, which shows seasonal and long-term trends. The day and year, on the other hand, stay close to 0 to ± 1, which suggests limited marginal influence. The model mainly uses spatial and sectoral information, with temporal features adding a little further detail. This shows that the learning behavior is steady and easy to understand. Figure 9 shows the SHAP explanation plot with samples arranged by how similar their attribution patterns are. This gives an overview of how feature contributions change from one observation to another. The x-axis shows about 2,800 samples that have been rearranged so that samples that are close to each other have similar SHAP value distributions. The y-axis shows the total SHAP contribution to the model output. Most of the samples have SHAP values that are stable and have low variance,

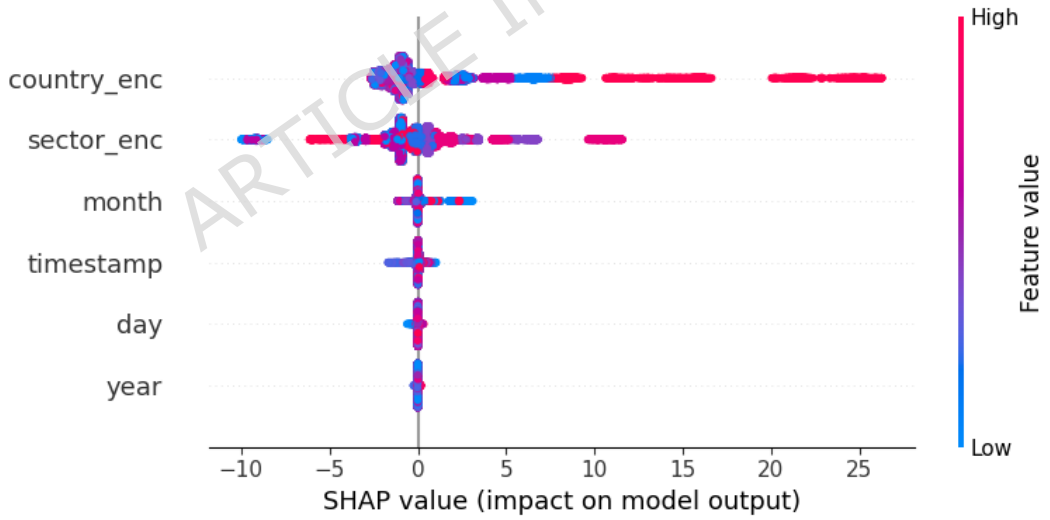


Figure 8. A global SHAP summary that shows how important each attribute is for predicting CO_2 emissions

which means they are mostly between -3 and $+5$. This shows that the model behaves the same way in most cases. On the other hand, a smaller group of samples, clearly evident in the rightmost area (samples 2,800–3,000), shows huge positive SHAP excursions of around +42, together with strong negative contributions of up to -5. This suggests high-impact emission regimes. The color bands (red to blue) show how different elements contribute to the overall representation. Sharp vertical transitions show changes in the structure of prominent features, like certain country-sector combinations or high-emission time periods. The smooth continuity across most areas shows strong generalization, while the grouped spikes show that there are specific emission patterns, not random noise. In general, this similarity-ordered visualization shows that the model captures coherent global structures, isolates high-emission outliers from baseline regimes, and keeps explainable consistency across the

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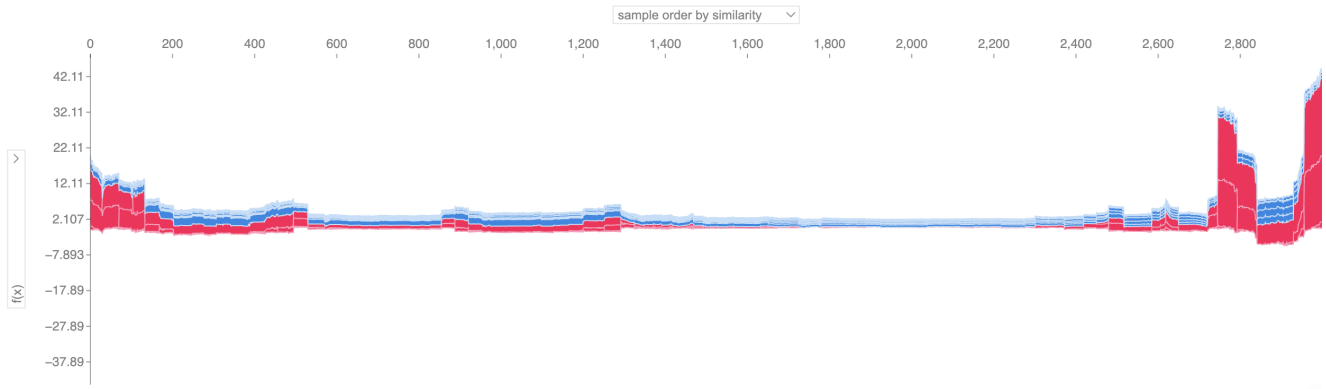


Figure 9. Global SHAP contributions are sorted by how similar the samples are, showing consistent emission patterns and clear high-impact regimes

dataset. This is a crucial trait for reliable CO_2 predictions. Figure 10 shows global SHAP attributions, with samples arranged by their estimated CO_2 outputs. In high-emission situations (around 32 to 42 units), SHAP contributions are very positive, reaching $+32$. This means that a small number of important features have a big effect on extreme CO_2 prediction. The SHAP magnitudes drop to about ± 5 when the output goes down to the medium range (about 2–10 units). This means that the feature effects are more evenly spread out. In the low-output domain, SHAP values are very close to zero ($|\text{SHAP}| < 2$), which means

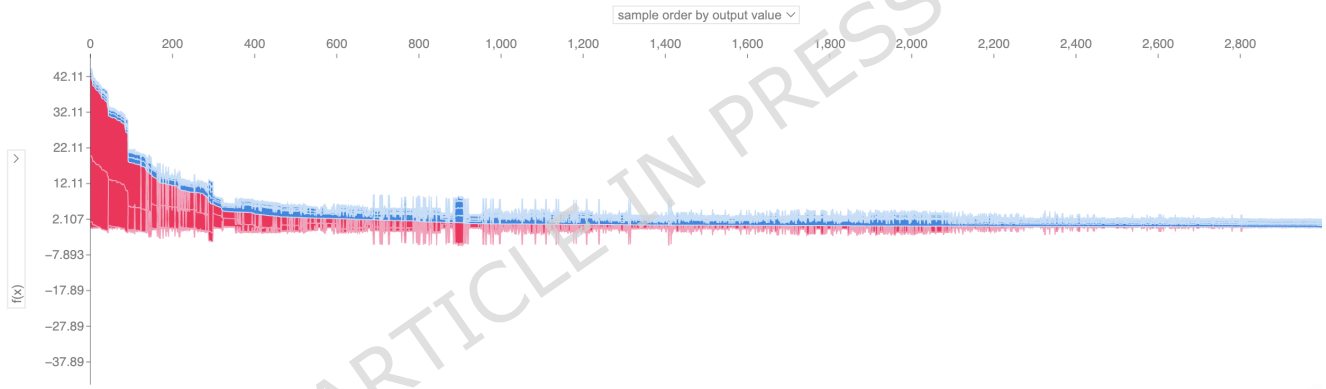


Figure 10. Global SHAP contribution plot with samples sorted by estimated CO_2 output, showing clear attribution tendencies

that the predictions are very close to the model baseline and that particular features cannot have much of an effect. The steady drop in SHAP magnitude across samples shows that the model is stable and that global interpretability is well-calibrated across all emission levels.

5 Conclusion

In this paper, we present an interpretable forecasting framework for CO_2 prediction based on BiLSTM and SHAP. The focus of this work is not to propose a new predictive architecture but to combine the predictive performance with explainability. The proposed framework accurately predicted the CO_2 time-series data by correctly capturing temporal dependencies. The results of the experiment showed that it worked well, with an RMSE of 8.58×10^5, an MAE of 1.19×10^5, a R^2 score of 0.99, and a low MAPE of 0.0073%. The results demonstrate strong predictive performance of the proposed model. In addition to enhancing predicted accuracy, SHAP made the model easier to understand on both a global and a local level. Global SHAP summary plots showed how important each input feature was compared to the others. Local explanations, on the other hand, used force and waterfall charts to give instance-level insights into each prediction. Ordering samples based on how similar their outputs were also helped us learn more about how consistent predictions are and how models behave. These explainability features make models more open and trustworthy, which is important for real-world applications like monitoring the environment and making decisions.

The proposed approach also can be scalable and adaptable to diverse environmental datasets. The forecasting framework may be appropriate to be used inside decision-support settings; however, a complete assessment of computing efficiency,

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inference delay and resource use was outside the scope of this study. These findings will be explored in a future effort to further assess the feasibility of real-time deployment. The integration of SHAP helps stakeholders find the main causes of CO_2 emissions, which makes it possible to make data-driven policies and tailored plans to reduce emissions. The proposed BiLSTM-SHAP framework achieves a successful balance between high prediction accuracy and explainable insights, making it a dependable and comprehensible approach for predicting CO_2 emissions in environmental time-series analysis.

Further studies may improve this framework by incorporating multivariate inputs, such as climate data, indications of industrial activity, energy consumption trends, urban traffic statistics, and socio-economic variables, to enhance the accuracy of predictive generalization. Studying sophisticated attention-based designs like Transformer models, hybrid CNN–BiLSTM models, and graph-based neural networks could help with modeling long-term dependencies and spatial-temporal correlations. Also, integrating the proposed model into real-time monitoring systems and testing it in other geographic areas, cities, and industries would make it more useful and reliable. Finally, using SHAP with causal explainability methodologies, counterfactual explanations, or reinforcement learning approaches may help us learn more about what really causes CO_2 emissions and make it easier to come up with targeted ways to reduce them. Also, combining the framework with decision-support dashboards and smart sensors that integrate with the Internet of Things (IoT) could help with ongoing monitoring, flexible forecasting, and proactive control of the environment.

Conflict of Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The code generated in the present work is with the corresponding author and may be supplied upon a reasonable request.

Author Contribution

D.T. Formal analysis, Experimentation. Methodology, Investigation, Writing – original draft. D.K. Formal analysis, Experimentation. Methodology, Investigation, Writing – original draft. S.S. Formal analysis, Methodology, Investigation, Writing – Review and Edit. M.A.W. Data Curation, Formal analysis, Result analysis, and Project administration. R.K.C. Formal analysis, Conceptualization, Result analysis, Writing – Review and Edit. J.B. Result analysis, Supervision, Conceptualization, Writing – Review and Edit. N.A.W. and M.M. Result analysis, Supervision, Writing – Review and Edit. P.Y. Result analysis, Supervision, Writing – Review and Edit.

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